31 March 1972
SUPERSEDING
MIL-R-45024C
21 November 1966

MILITARY SPECIFICATION

ROTARY TOOL KIT, ELECTRIC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type of portable, high speed rotary tool complete with accessories in carrying case.

1.2 Classification.- The rotary tool kit covered by this specifiation shall be of the following classes, as specified (see 6.2):

Class I - 18,000 RPM

Class II - 25,000 RPM

Class III - (see 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-T-7928 - Terminals, Lug and Splice, Crimp Style, Copper.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment).

FSC 5130

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MIL-W-17663 - Wheel, Abrasive, Mounted (Points).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1337 - Electric Portable Hand Tools General Supression System Design Requirements.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI)

B7.1 - 1970 Use, Care and Protection of Abrasive Wheels, Safety Code for the
B3.5 - Tolerances for Ball and Roller Bearings
B3.6 - Boundary Dimensions for Ball and Roller Bearings
B3.8 - Mounting Ball and Roller Bearings, Specification for

(Application for copies should be addressed to the American National Standards Institute, 10 East 40th Street, New York, New York 10016.)

National Bureau of Standards

Handbook H-28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Underwriters' Laboratories, Inc. Standards for Safety

UL 45 - Electric Tools

UL 498 - Attachment Plugs and Receptacles

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 1285 Walt Whitman Rd., Melville, L.I., New York, N. Y. 11746.)

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3. REQUIREMENTS

3.1 Standard product.- The rotary tool kit shall be the manufacturer's current standard product except for any changes necessary to comply with this specification.

3.2 Design.- The rotary tool kit shall be new. The design shall be such that the mounted points, wheels and rotary files can be utilized in the collet chuck of the rotary tool. All parts subject to wear, breakage or distortion shall be accessible for adjustment and repair. The rotary tool kit shall be complete so that when connected to the specified power source it can be used for the operation specified herein. The rotary tool kit shall be designed so that it can be utilized with a tool mount. The rotary tool sets shall be comprised principally of a housing, electric motor, chuck and a quill assembly and shall conform to table I for the specified class.

3.2.1 UL.- The rotary tool kit shall conform to Underwriters' Laboratory (UL) 45 standard, for use in nonhazardous locations.

3.2.1.1 Certification.- Prior to the approval of the first article or if none is submitted, prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the rotary tool kit he proposes to furnish under this specification meets the requirements specified in 3.2.1.

3.2.1.2 UL.- Acceptable evidence of meeting the requirements of Underwriters' Laboratories, Inc., shall be the UL label, or Listing Mark or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the rotary tool kits offered have been tested and conform to UL 45.

3.3 Materials.- Materials not definitely specified shall be of the quality normally used by the supplier for rotary tool kits, provided the completed item complies with all the provisions of this specification. Contact between dissimilar metals, especially brass, copper or steel in contact with aluminum or magnesium, shall be avoided wherever possible.

3.4 Construction.- Exposed surfaces of the rotary tool shall be constructed of materials that are inherently corrosion resistant, or of material protected against corrosion.

3.4.1 Castings.- All castings shall be uniform quality, free from blowholes, porosity, hard spots, shrinkage defects, cracks and other defects that may directly or indirectly affect the structural strength of the casting.

3.4.2 Welding, brazing and soldering.- Welding, brazing and soldering, when performed, shall be smooth, continuous and free of slag, flex and completely fused. In no event shall such processes, including peening or plugging, be used on castings or forgings for reclaiming any parts of the rotary tool kit.

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3.4.3 Fastening devices.- All screws, pins, bolts and similar parts shall be installed with means for preventing loss of tightness and adjustment. All parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 Electrical connections.- Electrical connections of conductors and terminal parts shall be of the pressure or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Flux for soldering shall be of a resin base. Crimp style terminals shall conform to MIL-T-7928.

3.4.5 Threads.- All threaded parts shall be in the inch system and shall comply with Handbook H28. Unified screw threads shall be utilized in lieu of American National screw threads and nonunified American Standard screw threads.

3.4.6 Lubrication.- All bearings (except sealed-for-life type) and other working parts shall be provided with means for lubrication.

3.4.7 Surfaces.- All castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled. External surfaces of the rotary tool shall be finished concentric with the spindle to allow accurate and rigid mounting when used with other basic machine tools.

TABLE I.- Classes and components

	Class I	Class II
Motor		
HP (Min.)	1/10	1/4
RPM (No load) (Min.)	18,000	25,000
RPM (Full load) (Min.)	8,000	14,000
Chuck		
Capacity	1/8 inch shank	1/8 inch and 1/4 inch shank
Runout (Max.)	0.003 inch	0.0015 inch

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TABLE I.- Classes and components (cont'd)

	Class I	Class II
Rotary tool		
Weight (Max.)	3 lbs.	4-1/2 lbs.
Overall length (Max.)	12 inches	13 inches
Body diameter (Max.)	3 inches	3 inches
Quill assembly length (Min.)	1-1/2 inches	2-1/2 inches
Abrasive wheels	Group 1 (see 3.4.9.6.1)	Group 2 & 3 (see 3.4.9.6.2 and 3.4.9.6.3)
Rotary files	Set - 1 (see 3.4.10.1)	Set - 2 (see 3.4.10.2)
Universal tool mount	-	One (when specified see 3.4.9.3 and 6.2)
Dressing stone	One (see 3.4.9.2)	One (see 3.4.9.2)
Tools	One set (see 3.4.9.4)	One set (see 3.4.9.4)
Flexible shaft	-	One (when specified see 3.4.9.1 and 6.2)
Spare parts		
Motor brushes	Six	Six
Air filters	-	-
Overload fuse	-	-
Carrying case	One (see 3.4.9.5)	One (see 3.4.9.5)
Net weight of complete set (Max.)	15 lbs.	35 lbs.

3.4.8 Components.- The rotor of the rotary tool shall be dynamically balanced to permit its operation without vibration (see 4.4.1).

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3.4.8.1 Housing.- The housing of the class I rotary tool shall be made of metal or plastic and shall not fracture or crack when tested as specified in 4.4.2. The material for the housing for the classes II and III shall be of the quality normally used by the supplier for rotary tool kits, and shall be provided with means for mounting on various basic machine tools with the use of a universal tool mount on various standard machine tools such as lathes, milling machines and drill presses.

3.4.8.2 Quill assembly.- The quill shall be made of commercial alloy steel, hardened and ground. The quill shall be mounted in prelubricated, sealed-for-life type ball bearings. A means shall be made for holding the spindle to facilitate the changings of tools. Length shall be as specified in table I.

3.4.8.3 Motor.- The rotary tool motor shall be universal and operate on single phase, 120 volts alternating current (AC), 50 to 60 Hertz, or 120 volts, direct current (DC). The motor shall have an intermittent duty rating at the horsepower and full load speed as specified with temperature rises as limited by UL 45.

3.4.8.4 Caution plate.- Motor housing shall incorporate a permanent warning in bold type, as a separate plate or as a portion of the standard label plate, reading "CAUTION" NO LOAD SPEED OF THIS TOOL IS _____ (actual speed inserted by manufacturer). DO NOT USE MOUNTED ABRASIVE WHEELS OR POINTS WHOSE SAFE OPERATING SPEED IS BELOW NO LOAD SPEED OF THIS TOOL."

3.4.8.5 Chuck.- The chuck shall be of the collet type, with a capacity and maximum allowable runout as specified in table I. The collet shall be of an alloy steel, hardened and ground to the tolerances necessary to meet the performance requirements of this specification. Runout of chuck shall be measured within 1 inch of the nose of the chuck.

3.4.8.6 Control.- An "on-off" switch shall be conveniently located on the rotary tool to turn the electrical current off and on. The switch shall conform to the requirements of UL 45.

3.4.8.7 Power cable and plug.- Rotary tools shall be furnished with not less than 8 feet of commercial type Underwriters' Laboratories approved three-conductor cable, as required for motor size furnished. Cable plug shall conform to UL 498 requirements for two-pole, 3-wire grounding devices rated 15 ampere, 125 volts.

3.4.9 Accessories.-

3.4.9.1 Flexible shaft.- A flexible shaft, not less than 36 inches in overall length, shall be furnished in accordance with table I. Means shall be provided for chucking the flexible shaft core in the collet check of the rotary tool. An adapter shall be provided at the drive end of the flexible

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shaft to secure the shaft casing to the housing of the rotary tool. The hand piece of the flexible shaft shall be not more than 1-1/2 inches in diameter and not less than 5 inches in length. The speed of the flexible shaft, when attached to the rotary tool, shall not be reduced more than 2000 RPM for the class II rotary tool. When bent in a "U" shape the RPM speed shall not be reduced more than 5000 RPM for the class II rotary tool (see 4.4.4).

3.4.9.2 Dressing stone.- One dressing stone made of silicon carbide shall be furnished in accordance with table I. The dressing stone shall be not less than 3 inches in length, 3/4 inches in width and 3/8 inches in thickness.

3.4.9.3 Universal tool mount.- When specified for class II (see 6.2), a tool mount shall be provided of the universal type, suitable for mounting the rotary tool on various standard basic machine tools such as a lathe, milling machine or drill press. The tool mount shall be provided with adjusting means that will permit the tool to be mounted at a height of at least 5 inches, measured from the center line of the spindle to the bottom of the base, and shall be capable of rigidly supporting the rotary tool at any position with this height. Universal tool mount shall be furnished in accordance with table I.

3.4.9.4 Tools.- One set of tools shall be furnished with each rotary tool (see table I). The tool set shall consist of a chuck wrench (if necessary) and any other tools necessary in the operation of the rotary tool.

3.4.9.5 Carrying case.- A carrying case shall be furnished in accordance with table I for each rotary tool. The carrying case shall be durable and of sufficient size to contain and orderly store the rotary tool and all accessories. The case shall be complete with hinged cover, carrying handle and closing fastener. The interior and exterior surfaces of the carrying case shall have a durable coating of paint or enamel, except surfaces which are inherently corrosion free or electroplated.

3.4.9.6 Abrasive wheels.- Abrasive wheels shall be furnished as indicated in table I. Groups shall be assembled as indicated herein. Each mounted wheel and mounted point provided with rotary tool kits shall have a safe operating speed (as cited in ANSI B7.1-1970) that is greater than the no-load speed of the grinder.

3.4.9.6.1 Group 1.- Group 1 wheels shall consist of one each of 26 standard abrasive wheels and points listed below. The abrasive wheels and points furnished shall conform to the requirements of MIL-W-17663 and shall embody the following characteristics:

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Abrasive	-	Aluminum oxide
Grit	-	As specified herein
Grade	-	0
Bond	-	Vitrified
Shape	-	As specified herein
Spindle	-	Diameter 1/8 inch, length suitable for 1/2 inch overhang with maximum engagement in chuck

Shape	Grit	Shape	Grit	Shape	Grit	Shape	Grit
B-42	60	B-73	80	W-144	60	W-176	60
B-44	60	B-103	80	W-152	60	W-183	60
B-51	60	B-112	60	W-154	60	W-185	60
B-52	90	B-121	60	W-160	60	W-193	60
B-53	90	B-131	60	W-163	60	W-200	60
B-63	60	B-135	60	W-173	80		
B-70	60	W-143	60	W-175	60		

3.4.9.6.2 Group 2.- Group 2 wheels shall consist of one each of 20 standard abrasive wheels and points listed below. The abrasive wheels and points furnished shall conform to the requirements of MIL-W-17663 and shall have the following characteristics:

Abrasive	-	Aluminum oxide
Grade	-	0
Bond	-	Resinoid or vitrified
Spindle	-	Diameter 1/8 or 1/4 inch (see table I), length suitable for 1/2 overhang with maximum engagement in chuck

Wheels and points shall be of the following shapes and grit:

Shape	Grit	Shape	Grit
		B-73	80
		B-97	90
A-22	60	B-103	80
A-24	90	B-106	90
		B-114	90
A-26	60	B-115	90
		B-122	80
B-81	90	B-123	90
		B-124	90
		B-132	80
B-45	90	B-136	90
B-52	90	W-173	80
B-53	90	W-201	60

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3.4.9.6.3 Group 3, abrasive wheels, resilient.- Group 3 abrasive wheels shall consist of one each of the 12 resilient wheel listed below. Group 3 wheels shall comply with MIL-W-17663.

Abrasive	- Aluminum oxide
Bond	- Resinoid (with cotton fiber) or vitrified
Spindle	- 1/8 or 1/4 inch diameter, length suitable dimensions for 1/2 inch overhang with maximum engagement in chuck

Shape	Grit
*A-5	80
A-12	80
*A-22	80
*B-41	80
B-51	80
B-52	80
*B-61	80
*B-91	80
B-102	80
*W-186	80
*W-202	80
*W-203	80

NOTE * 1/4" diameter shank only shall be acceptable.

3.4.10 Rotary files.- When specified (see 6.2), rotary files shall be furnished in the following sets and shall be in accordance with table I.

3.4.10.1 Set 1, rotary files.- Set 1 rotary files shall consist of one each of eight files listed in table II. Set 1 files shall be high speed steel with 1/8 inch diameter shanks.

TABLE II.- Set 1, rotary files

Shape	Maximum diameter	Length cut	Cut
	Inch	Inch	
Cylinder	1/4	1/2	Standard
Cylinder (radius end)	1/4	1/2	Standard
Tree	1/4	1/2	Standard
Tree, pointed	1/4	1/2	Standard
Cone	1/4	1/2	Standard
Ball	1/4	1/4	Standard
Oval	1/4	3/8	Standard
Inverted cone	1/4	1/4	Standard

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3.4.10.2 Set 2, rotary files.- Set 2 rotary files shall consist of the 20 files listed in table III. The files shall be made of high speed steel with 1/4 inch diameter shanks and length for 1/2 inch overhang with maximum engagement in chuck. The shape and dimensions of the cutting head and the pitch of the cutting teeth shall be as specified in table III.

TABLE III.- Set 2 rotary files and cutting heads

Shape	Dimensions		Cutting teeth (minimum)
	Maximum diameter	Length	
	Inch	Inches	(Per Inch)
Cylindrical	3/16	1/2	20
Cylindrical	1/4	1	25
Cylindrical	3/8	1	20
Cylindrical (radius end)	1/8	1/2	50
Cylindrical (radius end)	1/4	1	25
Cylindrical (radius end)	3/8	1	20
Ball	1/8	1/8	62
Ball	1/4	1/4	62
Ball	3/16	3/16	62
Olive (or oval)	3/8	5/8	25
Olive (or oval)	1/2	7/8	20
Taper (7 degrees)	3/8	1	31
Taper (30 degrees)	3/8	3/8	31
Tree (radius end)	1/2	1	20
Tree (pointed)	1/4	3/4	40
Tree (pointed)	3/8	3/4	31
Tree (pointed)	1/2	1-1/8	20
Inverted cone	1/2	1/2	20
Bi-shape (convex concave)	5/16	1-1/8	25
Concave radius	1/2	3/8	25

3.5 Electromagnetic interference suppression.- When specified (see 6.2), rotary tools shall comply with the class I requirements of MIL-STD-1337.

3.6 Fungus proofing.- When specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose function will be adversely effected by the varnish coating shall not be treated.

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When used, the varnish shall be applied to give a minimum dry film thickness of 0.001 inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.7 Endurance.- The rotary tool shall be operated under no load for a period of not less than 2 hours. The RPM speed after each two periods shall not be less than the speed specified for the indicated class and, in addition, shall not be damaged in any way when tested as specified in 4.4.6.

3.8 Finish.- The rotary tool and carrying case shall be finished in accordance with the manufacturer's standard commercial practice.

3.9 Workmanship.- The rotary tool and case shall be free of burrs, slivers, grit, protrusions and dents. The finish shall be smooth, free of embedded foreign matter, sags, runs and areas of no film.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of reference specifications, drawings and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Intermediate test of rotor.- The rotors of the rotary tool shall be tested for vibration in accordance with 4.4.1. The lot shall be all the rotors offered for testing at one time. The sample unit shall be one rotor. The inspection level shall be S-2 and the acceptable quality level shall be 4.0, expressed in defects per hundred units.

4.2.3 End item inspection.- The lot shall consist of all rotary tool kits of one class offered for inspection at one time. The sample unit shall be one completely finished rotary tool kit.

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4.2.3.1 Visual inspection.- The rotary tool kits shall be examined for defects classified in table IV. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE IV.- Classification of defects

Examination	Defect	Classification Major	Minor
Finish	Not manufacturer's standard commercial finish		X
	Not smooth, free of embedded foreign matter, sags, runs and areas of no film		X
Construction and workmanship	Materials not protected for corrosion resistant, not free of burrs, slivers, protrusions, dents	X	
	Castings not uniform, free of blow-holes, porosity, hard spots, shrinkage and cracks	X	
	Lubrication means not provided where required	X	
	Component missing	X	
	Component not as specified	X	
	Off-on switch not as specified	X	
Welding, brazing soldering	Not smooth, continuous and free of slag, flux and fusion incomplete	X	
Electrical connections	Not pressure or solder type	X	
	Not mechanically secure before soldering		X
Fastening devices	Screws, pins, bolts not installed with a means to prevent loss of tightness	X	
	Parts subject to removal swaged, peened or staked	X	
Molded forged stampings	Not free of sand, dirt, fins, sprues, scale and flux	X	
	Edges not rounded or beveled		X
Wheel sets and rotary files sets	Missing, incomplete	X	

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4.2.3.2 Dimensional examination.- Inspection shall be made of the end item for compliance with dimensions specified. Any nonconformance shall constitute a defect. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

4.2.3.3 End item testing.- Rotary tool kits shall be tested as specified in 4.4.2 through 4.4.6. The inspection level shall be S-2. Failure to pass any test shall be cause to reject the lot.

4.3 Preparation for delivery.- An examination of preparation for delivery shall be made in accordance with PPP-P-40.

4.4 Tests.-

4.4.1 Balancing test.- Rotary moving components shall be tested dynamically at maximum speeds by any commercial method. Any indication of abnormal vibration shall constitute failure of this test.

4.4.2 Drop test.- A completely assembled rotary tool less a mounted wheel or rotary file shall be dropped from a height of 2 feet not less than three times onto a hard wood floor. The tool shall not be dropped on the quill end. Any evidence of fracture, crack or failure to operate shall constitute failure of this test.

4.4.3 Chuck runout test.- A test mandrel with combined error not exceeding 0.0001 inch in straightness and out of roundness, shall be used for the test. Nominal dimensions of the mandrel shall be 1/8 inch in diameter for class I and 1/4 inch for class II. Length shall be 2 inches. The mandrel shall be secured in the collet chuck at the rotary tool. The rotary tool shall be clamped in a cushioned vise tight enough to assure stability. While the test mandrel is turned slowly by hand, an indicator reading shall be taken on the mandrel within 1 inch of the nose of the chuck. Runout exceeding 0.003 inch TIR (total indicator runout) for class I and 0.0015 inch for class II shall constitut failure of this test.

4.4.4 Speed reduction test (class II).- The class II rotary tool shall be operated at no load and quill speed shall be measured by tachometer and noted. The flexible shaft shall then be connected to the rotary tool and shall be held by the handpiece in such a manner as to keep the shaft straight within plus or minus 1 inch. While the shaft is being held in this manner, a tachometer shall be used to determine no load quill speed. The test shall be repeated with the shaft bent to a U-shape with the flexible shaft handpiece parallel with and within 12 inches of the rotary tool. The quill speeds with the flexible shaft

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attached, as specified above, shall be compared with quill speed noted before flexible shaft attachment. With the shaft straight, the quill speed shall not be reduced by more than 2000 RPM. With the shaft bent into a U-shape, the speed shall be reduced not more than 5000 RPM. Any noncompliance shall constitute failure of the test.

4.4.5 Electromagnetic interference suppression test.- When specified (see 6.2), tests for electromagnetic interference suppression shall be conducted on electrical motors in accordance with class I of MIL-STD-1337.

4.4.6 Endurance test.- The rotary tool shall be clamped in a cushioned vise and allowed to run with no load for a period of not less than 2 hours. At the end of the 2 hour period, the rotary tool shall be examined for evidence of damage and speed shall be checked to see that the rotary tool is up to rated speed. Evidence of damage or speeds less than specified no load speeds shall constitute failure of this test.

4.5 Inspection of preservation, packaging, packing and marking for shipment and storage.- Inspection of preservation, packaging, packing and marking for shipment and storage shall be in accordance with PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing shall be in accordance with PPP-P-40 for Grinders, Electric, Portable. Levels of preservation, packaging and packing shall be as specified (see 6.2).

5.2 Marking.- Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The rotary tool kit described in this specification is intended for use in operations such as deburring, filing, light milling, grinding and polishing. The mounted wheels and points supplied with the kit can be used on any metal from nonferrous alloys to hardened steel. The rotary files can be used on nonferrous metals and alloys and on plastics and wood. When used in conjunction with a standard machine tool, the rotary tool shall be capable of performing precision work in external, internal and form grinding operations.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class required (see 1.2).

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(c) Flexible shaft, when required (see table I and 3.5).
(d) Tool mount when required (see table I).
(e) When abrasive wheels groups are not to be assembled as indicated by groups 1, 2 or 3 (see 3.4.9.7).
(f) Level of preservation and packaging, and level of packing required (see 5.1).
(g) When electromagnetic interference suppression is required (see 3.5).
(h) When fungus treatment is required (see 3.6).

6.3 Contract data requirements.- Data conforming to Data Item Description DI-M-4022 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.4 Deletion of requirements.- The requirements for the class III rotary tool kit have been deleted from this specification since there are no known requirements.

Custodians:

Army - GL
Air Force - 84

Preparing activity:

Army - GL

Project No. 5130-0214

User activities:

Army - ME
Navy - MC, SH

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R25

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in th use of the specification in procurement of products for ultimate use by the Department of Defense. This shee is provided for obtaining information on the use of this specification which will insure that suitable product can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Commen and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Rotary Tool Kit, Electric MIL-R-45024D

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Natick, Massachusetts 01760

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